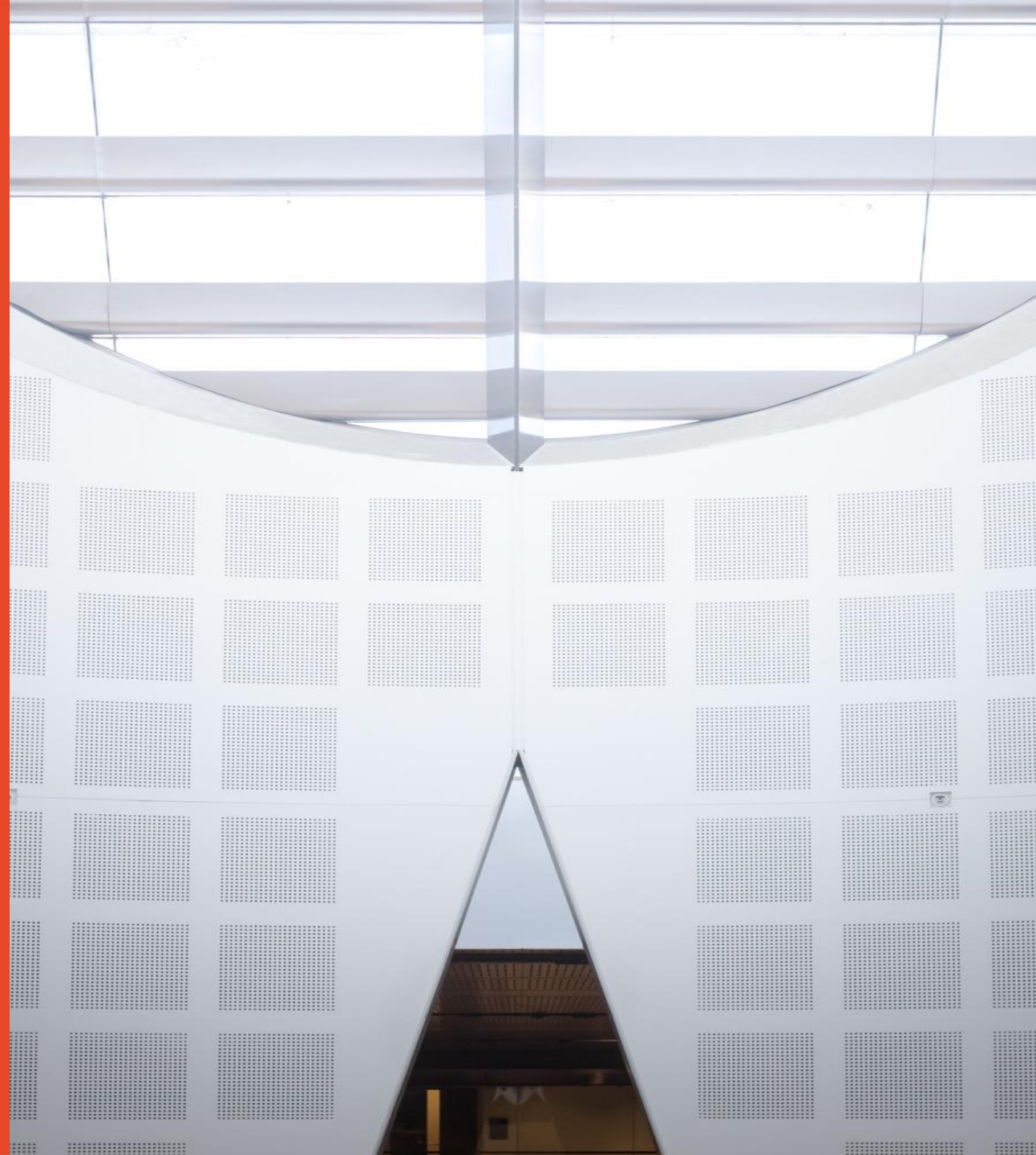


Semiconductor Physics

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Non Equilibrium

Non-equilibrium in semiconductors occurs when external perturbations disturb the balance between carrier

Carrier concentrations (electrons and holes) deviate from their thermodynamic equilibrium state

Deviation from Equilibrium:

In a non-equilibrium state, this stable, uniform distribution is disturbed, leading to a non-uniform distribution or an excess of carriers

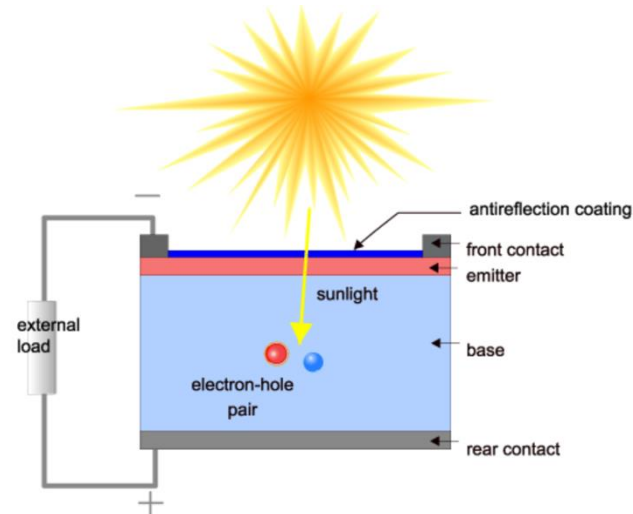
Quasi-Fermi Levels:

To account for the non-equilibrium carrier distributions, the concept of quasi-Fermi levels is introduced. These levels allow for the separate mathematical description of electron and hole populations, even when the overall semiconductor is not in equilibrium

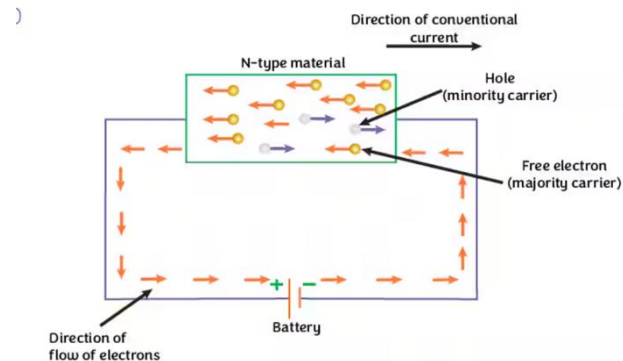


Non Equilibrium

Light Illumination: External light can generate electron-hole pairs, increasing their concentration beyond the equilibrium levels.

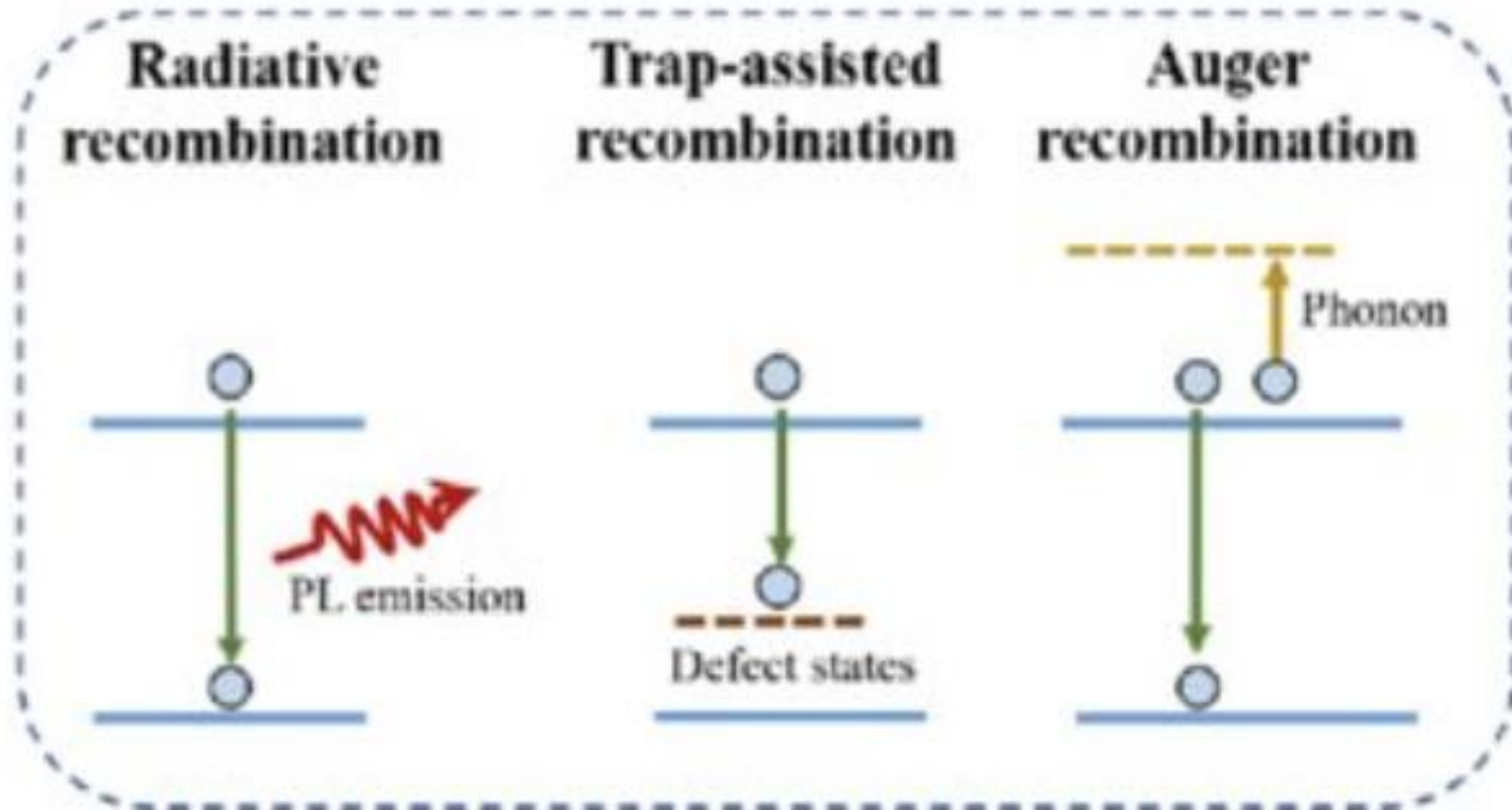


External Electric Fields: The application of an electric field causes carriers to move, resulting in drift and diffusion currents that can create non-equilibrium spatial distributions of carriers.



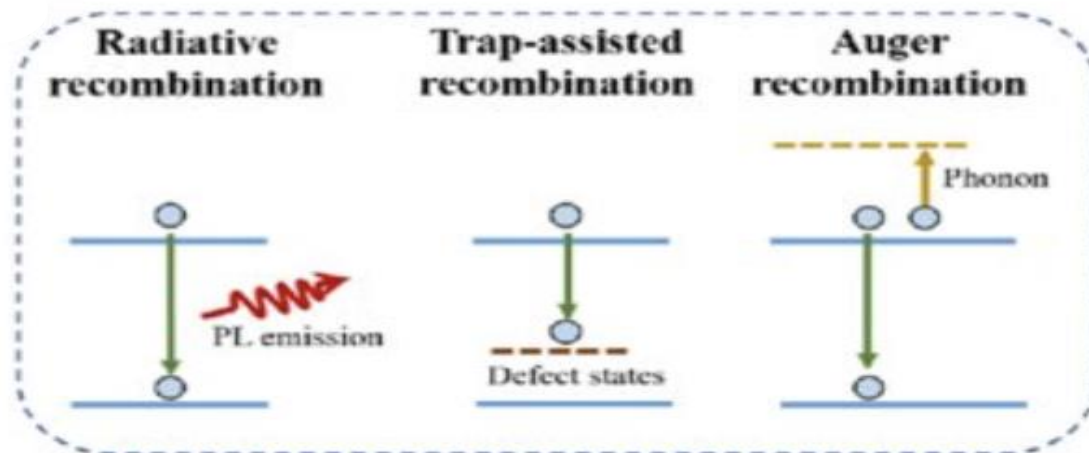
CARRIER GENERATION AND RECOMBINATION

Three types of Generation-Recombination processes



Recombination Mechanisms

Mechanism	Type	Energy released as	Dominant when
Radiative	Photon emission	Light	Clean, direct-gap semiconductors (e.g. GaAs)
SRH (trap-assisted)	Phonon emission	Heat	Defective, indirect-gap semiconductors (e.g. Si)
Auger	Carrier excitation + phonons	Heat	High injection or heavy doping



CARRIER GENERATION AND RECOMBINATION

Generation is the process whereby electrons and holes are created, and

Recombination is the process whereby electrons and holes are annihilated.

Since the net carrier concentrations are independent of time in thermal equilibrium, the rate at which electrons and holes are generated and the rate at which they recombine must be equal.

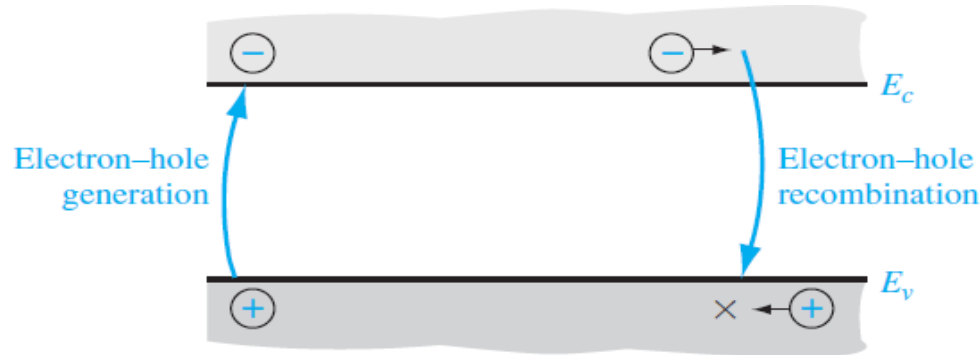


Figure 6.1 | Electron-hole generation and recombination.

The general law:

$$\frac{dn}{dt} = G - R$$

At steady state, $G = R$

In non-equilibrium, $\Delta n = n - n_0 \neq 0$, where n is the equilibrium carrier concentration.

CARRIER GENERATION AND RECOMBINATION

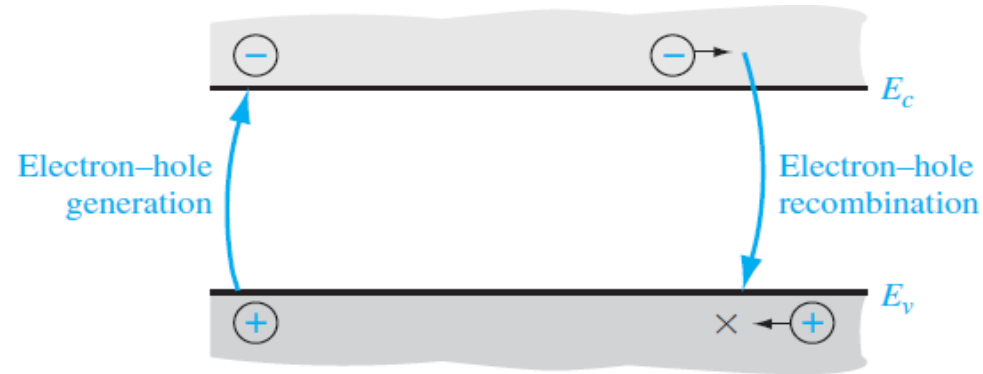


Figure 6.1 | Electron-hole generation and recombination.

At equilibrium

Consider direct band-to-band generation and recombination

Let G_{n0} and G_{p0} be the thermal-generation rates of electrons and holes, respectively.

For the direct band-to-band generation, the electrons and holes are created and recombine in pairs, so we must have that

$$R_{n0} = R_{p0} \quad G_{n0} = G_{p0}$$

In thermal equilibrium, the **concentrations of electrons and holes are independent of time**; therefore, the generation and recombination rates are equal, so we have

$$G_{n0} = G_{p0} = R_{n0} = R_{p0}$$

Excess Carrier Generation and Recombination

non-equilibrium

The additional electrons and holes created are called excess electrons and excess holes.

The excess electrons and holes are generated by an external force at a particular rate.

Let g_n be the generation rate of excess electrons and g_p be that of excess holes. These generation rates also have units of $\#/cm^3s$.

$$g'_n = g'_p$$

$$n = n_0 + \delta n$$

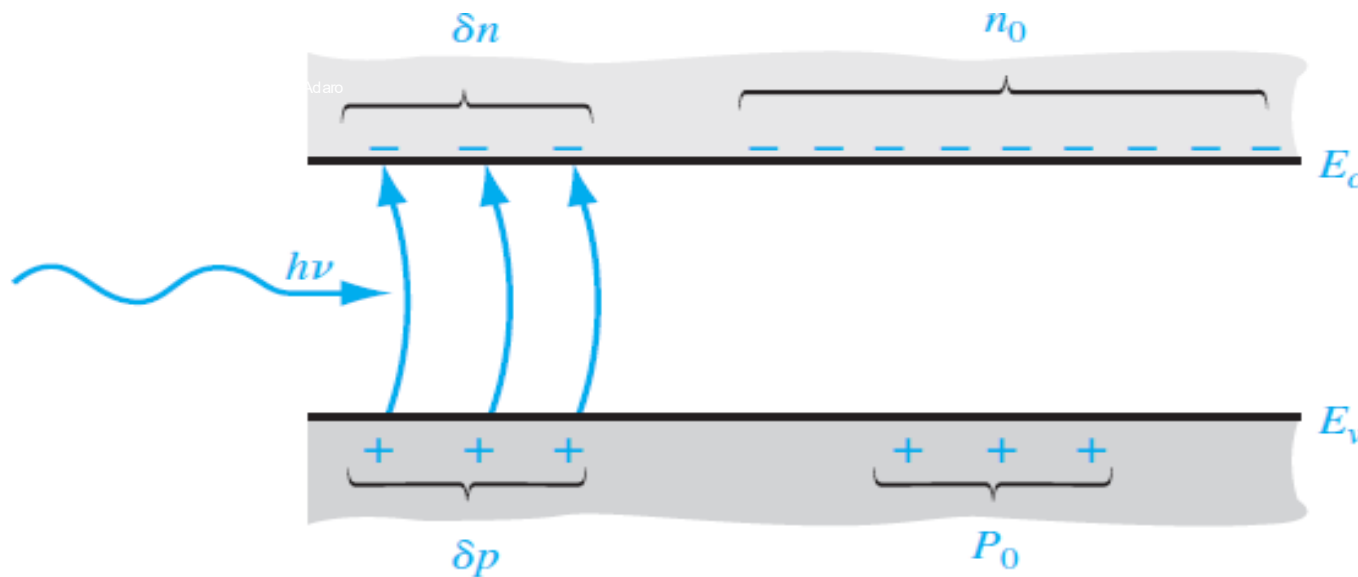
$$p = p_0 + \delta p$$

where n_0 and p_0 are the thermal-equilibrium concentrations,

and δn and δp are the excess electron and hole concentrations.

Excess Carrier Generation and Recombination

Figure 6.2 shows the excess electron–hole generation process and the resulting carrier concentrations. The external force has perturbed the equilibrium condition so that the semiconductor is no longer in thermal equilibrium.



Non-equilibrium condition

$$np \neq n_i^2$$

Figure 6.2 | Creation of excess electron and hole densities by photons.

Excess Carrier Generation and Recombination

- A steady-state generation of excess electrons and holes will not cause a continual buildup of the carrier concentrations.
- As in the case of thermal equilibrium, an electron in the conduction band may “fall down” into the valence band, leading to the process of excess electron–hole recombination.

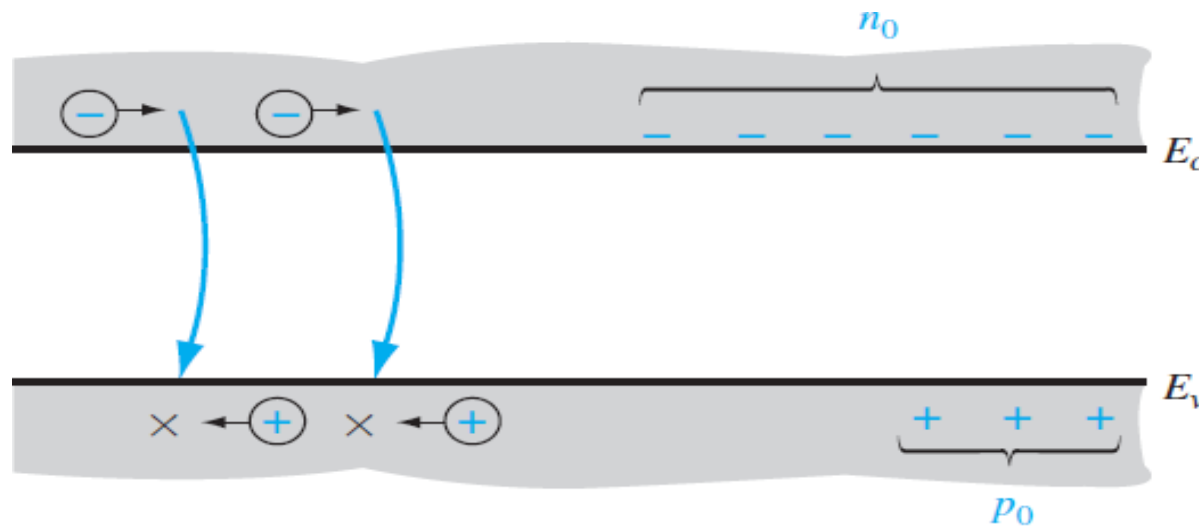


Figure 6.3 | Recombination of excess carriers reestablishing thermal equilibrium.

Excess Carrier Generation and Recombination

The recombination rate for excess electrons is denoted by R_n and for excess holes by R_p . Both parameters have units of $\#/cm^3s$.

The excess electrons and holes recombine in pairs, so the recombination rates must be equal. We can then write

$$R_n = R_p$$

In the direct band-to-band recombination that we are considering, the recombination occurs spontaneously; thus, **the probability of an electron and hole recombining is constant with time.**

Low level injection limit

Primary condition

$$\Delta n = \Delta p$$

in **p-type semiconductor** where the **majority carriers** are holes

Low-level injection means:

$$\Delta n = \Delta p \ll p_0$$

- But since the **excess is tiny compared to** p_0 (the doping density), we can safely say:

$p \approx p_0$ (the majority carrier concentration is almost unchanged)

$n = n_0 + \Delta n$ (minority carriers change significantly)

Low level injection limit

- The **thermal equilibrium generation rate** is given by:

$$G_0 = \alpha_r n_i^2$$

where α_r is the recombination coefficient and n_i is the intrinsic carrier concentration.

- In equilibrium, the rate of generation equals the rate of recombination.
- When the system is **perturbed** (illumination, injection), additional carriers appear:

$$n = n_0 + \delta n(t), \quad p = p_0 + \delta p(t)$$

Because electrons and holes are created in pairs:

$$\delta n(t) = \delta p(t)$$

Low level injection limit

$$\frac{d(\Delta n)}{dt} = \alpha_r [n_i^2 - (n_0 + \Delta n)(p_0 + \Delta p)]$$

$$\frac{d\Delta n}{dt} = -\alpha_r \Delta n [(n_0 + p_0) + \Delta n]$$

Under **p-type low-level injection**, the excess term Δn is negligible compared to p_0 , and since $p_0 \gg n_0$:

$$\frac{d\Delta n(t)}{dt} \approx -\alpha_r \Delta n(t) p_0$$

$$\Delta n(t) = \Delta n(0) e^{-\alpha_r p_0 t} = \Delta n(0) e^{-t/\tau_{n0}}$$

where:

$$\tau_{n0} = \frac{1}{\alpha_r p_0}$$

A larger p_0 (more heavily doped material) → **shorter lifetime**, because recombination events are more frequent.

Excess Carrier Generation and Recombination

The recombination rate of the majority carrier electrons will be the same as that of the minority carrier holes, so we have

$$R_n = R_p = \frac{\Delta n(t)}{\tau_{n0}}$$

The generation rates of excess carriers are not functions of electron or hole concentrations. In general, the generation and recombination rates may be functions of the space coordinates and time.

Characteristics of Excess Carriers

$$F_{px}^+(x + dx) = F_{px}^+(x) + \frac{\partial F_{px}^+}{\partial x} \cdot dx$$

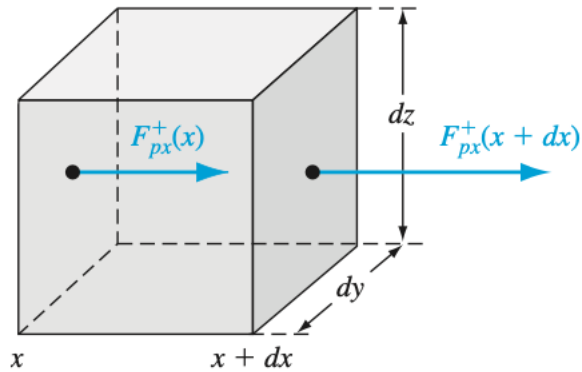


Figure 6.4 | Differential volume showing x component of the hole-particle flux.

increase in the number of carriers per unit time within the differential volume element due to the x-component of hole flux is given by

$$\frac{\partial p}{\partial t} dx dy dz = [F_{px}^+(x) - F_{px}^+(x + dx)] dy dz = -\frac{\partial F_{px}^+}{\partial x} dx dy dz$$

The net increase in the number of holes per unit time in the differential volume element is then given by

$$\frac{\partial p}{\partial t} dx dy dz = -\frac{\partial F_p^+}{\partial x} dx dy dz + g_p dx dy dz - \frac{p}{\tau_{pt}} dx dy dz$$

$$\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$$

$$\frac{\partial n}{\partial t} = -\frac{\partial F_n^-}{\partial x} + g_n - \frac{n}{\tau_{nt}}$$

Time-Dependent Diffusion Equations

$$J_p = e\mu_p pE - eD_p \frac{\partial p}{\partial x}$$



$$D_p \frac{\partial^2 p}{\partial x^2} - \mu_p \left(E \frac{\partial p}{\partial x} + p \frac{\partial E}{\partial x} \right) + g_p - \frac{p}{\tau_{pt}} = \frac{\partial p}{\partial t}$$

$$J_n = e\mu_n nE + eD_n \frac{\partial n}{\partial x}$$

$$D_n \frac{\partial^2 n}{\partial x^2} + \mu_n \left(E \frac{\partial n}{\partial x} + n \frac{\partial E}{\partial x} \right) + g_n - \frac{n}{\tau_{nt}} = \frac{\partial n}{\partial t}$$

At Steady State

At steady state ($\partial/\partial t = 0$):

$$D_n \frac{\partial^2 n}{\partial x^2} - \frac{n}{\tau_{nt}} = 0$$

Excess Carrier Generation and Recombination: Temporal decay

Equation (low-level injection) = equal number of excess carrier $\Delta n = \Delta p$

$$\frac{d \Delta n(t)}{dt} = -\frac{\Delta n(t)}{\tau}$$

Solution:

$$\Delta n(t) = \Delta n(0) e^{-t/\tau}$$

- τ depends on the dominant recombination process (SRH, radiative, Auger):

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_{\text{SRH}}} + \frac{1}{\tau_{\text{rad}}} + \frac{1}{\tau_{\text{Auger}}}$$

Excess Carrier Generation and Recombination

$$D \frac{d^2 \Delta n(x)}{dx^2} - \frac{\Delta n(x)}{\tau} = 0$$

Solution for a semi-infinite region (surface)

$$\Delta n(x) = \Delta n(0) e^{-x/L}$$

$$L = \sqrt{D\tau}$$

The injected excess carriers decay exponentially as they move into the material, with a characteristic distance L.

Quasi Fermi Levels

The thermal-equilibrium electron and hole concentrations are functions of the Fermi energy level

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right)$$

$$p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

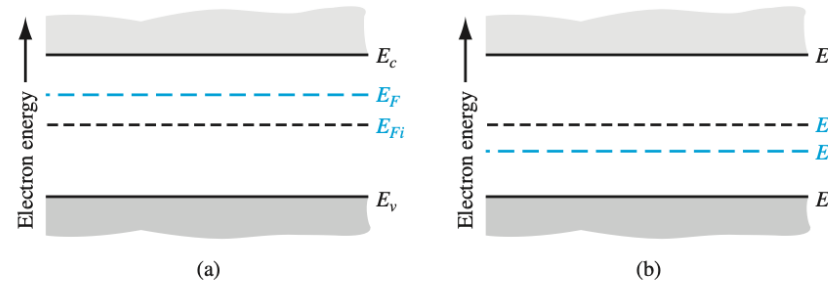


Figure 6.14 | Thermal-equilibrium energy-band diagrams for (a) n-type semiconductor and (b) p-type semiconductor.

In non-equilibrium The quasi-Fermi energy level for the minority carrier holes is significantly different from the Fermi level and illustrates the fact that we have deviated from thermal equilibrium significantly

$$n_0 + \delta n = n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right)$$

$$p_0 + \delta p = n_i \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$$

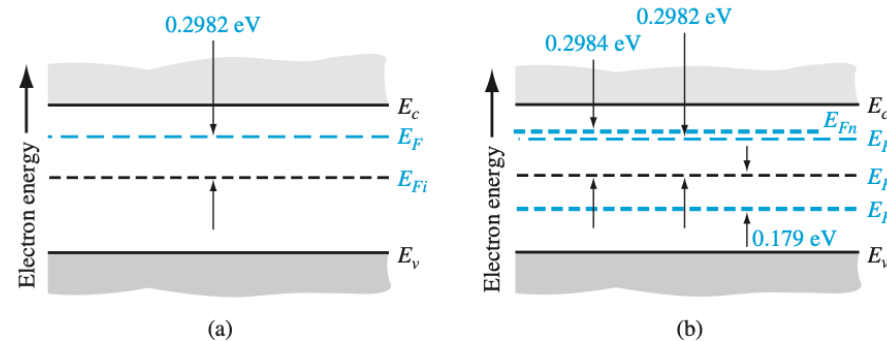


Figure 6.15 | (a) Thermal-equilibrium energy-band diagram for $N_d = 10^{15} \text{ cm}^{-3}$ and $n_i = 10^{10} \text{ cm}^{-3}$. (b) Quasi-Fermi levels for electrons and holes if 10^{13} cm^{-3} excess carriers are present.

In p-type semiconductors:

- Under low injection $\rightarrow E_{Fn}$ moves up, E_{Fp} stays put
- Under high injection $\rightarrow$ both E_{Fn} and E_{Fp} move symmetrically away from E_i

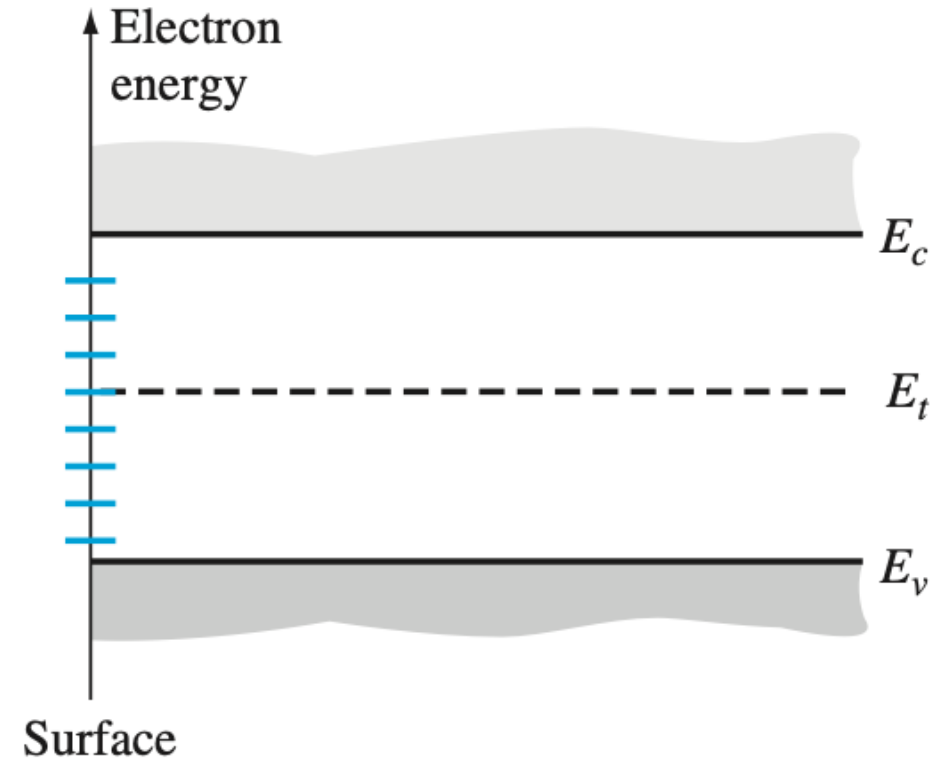
Shockley–Read–Hall Theory of Recombination

In a real semiconductor material, defects occur within the crystal and disrupt the perfect periodic-potential function.

An allowed energy state, also called a trap, within the forbidden bandgap may act as a recombination center, capturing both electrons and holes with almost equal probability.

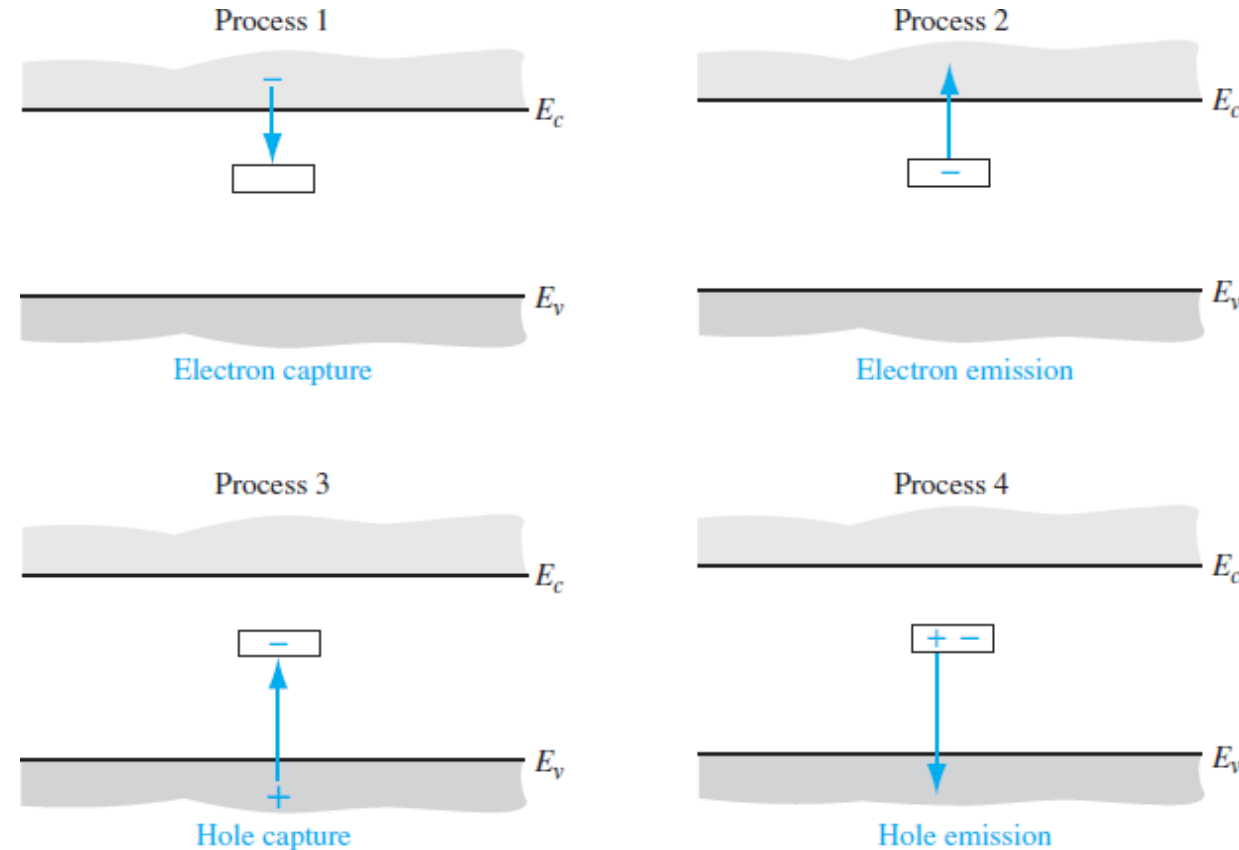
This equal probability of capture means that the capture cross sections for electrons and holes are approximately equal.

The Shockley–Read–Hall theory of recombination assumes that a single recombination center, or trap exists at an energy E_t within the bandgap.



Shockley–Read–Hall Theory of Recombination

- Process 1: The capture of an electron from the conduction band by an initially neutral empty trap.
- Process 2: The inverse of process 1 — the emission of an electron that is initially occupying a trap level back into the conduction band.
- Process 3: The capture of a hole from the valence band by a trap containing an electron. (Or we may consider the process to be the emission of an electron from the trap into the valence band.)
- Process 4: The inverse of process 3 — the emission of a hole from a neutral trap into the valence band. (Or we may consider this process to be the capture of an electron from the valence band.)



Shockley–Read–Hall Theory of Recombination

Assuming p-type extrinsic semiconductor: Electron and hole capture rate ($\text{cm}^{-3}\cdot\text{s}^{-1}$)

The two capture rates still have the same general forms:

$$R_{cn} = C_n N_t (1 - f_F(E_t)) n, \quad R_{cp} = C_p N_t f_F(E_t) p.$$

R_{cn}	Electron capture rate ($\text{cm}^{-3}\cdot\text{s}^{-1}$)
C_n	Capture coefficient = $\sigma_n v_{th}$, proportional to capture cross-section and carrier thermal velocity
N_t	Density of traps (cm^{-3})
$f_F(E_t)$	Fermi-Dirac probability that the trap is already occupied by an electron

So this term describes **how fast electrons fall into the traps**.

Shockley–Read–Hall Low level injection

The same logic will later be applied to **hole capture** from the valence band, leading to the full SRH recombination rate formula:

complete SRH rate equation

$$U_{SRH} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

$$n_1 = n_i e^{(E_t - E_i)/kT}, \quad p_1 = n_i e^{(E_i - E_t)/kT}$$

n_1 and p_1 effective carrier concentrations when the **trap level**

In a p-type material ($p_0 \gg n_0$):

$$U_{SRH} \approx \frac{p_0 \Delta n}{\tau_p(n_0 + n_1) + \tau_n(p_0 + p_1)}$$

If $p_0 \gg n_1$ (which is common), this simplifies to:

$$U_{SRH} \approx \frac{\Delta n}{\tau_n(p_0 + p_1)/(p_0 + n_1)}$$

the lifetime is inversely proportional to **trap density** and **capture efficiency**.

$$\tau_{eff} = \frac{\tau_p(n_0 + n_1) + \tau_n(p_0 + p_1)}{p_0 + n_0 + n_1 + p_1}.$$

- σ : capture cross-section of the trap (cm^2)
- v_{th} : thermal velocity of carriers (cm/s)
- N_t : trap density (cm^{-3})

Shockley–Read–Hall High level injection

The same logic will later be applied to **hole capture** from the valence band, leading to the full SRH recombination rate formula:

complete SRH rate equation

$$U_{SRH} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)} \quad n_1 = n_i e^{(E_t - E_i)/kT}, \quad p_1 = n_i e^{(E_i - E_t)/kT}$$

n_1 and p_1 effective carrier concentrations when the **trap level**

In high-level injection:

$$n \approx p \approx \Delta n \quad \text{and} \quad \Delta n \gg n_1, p_1$$

$$U_{SRH} \approx \frac{(\Delta n)^2 - n_i^2}{(\tau_p + \tau_n)\Delta n}$$

$$\tau_{eff} = \frac{\tau_p(n_0 + n_1) + \tau_n(p_0 + p_1)}{p_0 + n_0 + n_1 + p_1}.$$

Since $\Delta n \gg n_i$:

$$U_{SRH} \approx \frac{\Delta n}{\tau_p + \tau_n}$$

$$\tau_{eff} = \frac{\Delta n}{U_{SRH}} \approx \tau_p + \tau_n$$

- It no longer depends on the trap energy E_t or on doping level.

Shockley–Read–Hall Theory of Recombination

Parameter	Effect on Recombination
Trap density (N_t) ↑	Recombination increases (lifetime ↓)
Trap at midgap	Maximum recombination rate
Trap near band edges	Recombination slower
Higher temperature	Faster emission from traps → reduced net recombination

Auger recombination Process

Auger recombination is a **non-radiative three-particle process**.

It occurs when an **electron–hole pair recombines**, but instead of releasing a photon (as in radiative recombination), the **released energy is transferred to a third carrier**—another electron or hole—which becomes highly energetic (“hot”) and then thermalises by losing energy to the lattice as phonons.

So, energy is **not emitted as light but** dissipated as **heat**.

Two main channels exist:

1.eeh process:

2. Two electrons and one hole participate.

3. One electron–hole pair recombines; the excess energy excites the *other* electron to a higher conduction-band state.

4.ehh process:

5. Two holes and one electron participate.

6. One e–h pair recombines; the released energy is transferred to another hole, sending it deeper into the valence band.

Auger recombination Process

It's a **non-radiative three-particle process**. Does not produce light, only heat

Both are described generically by:

$$R_{Auger} = C_n n^2 p + C_p n p^2$$

where:

- C_n, C_p = Auger recombination coefficients (cm⁶/s)
- n, p = electron and hole concentrations

Two mechanisms:

- **eeh** → the energy is given to another **electron** → term $C_n n^2 p$
- **ehh** → the energy is given to another **hole** → term $C_p n p^2$

Auger recombination Process

Low level injection

The rate becomes:

In a p-type semiconductor, $p_0 \gg n_0$

$$p \approx p_0, \quad n = n_0 + \Delta n, \quad \Delta n \ll p_0$$

$$R_{Auger} \approx C_p p_0^2 \Delta n$$

- Recombination is **linear** with the number of excess carriers Δn .
- It depends strongly on the **majority carrier concentration** (p_0^2).
- Dominates in heavily doped materials because of the p_0^2 factor.

The corresponding **minority-carrier lifetime** is:

$$\tau_{Auger} = \frac{1}{C_p p_0^2}$$

Auger recombination Process

High level injection

$$n \approx p \approx \Delta n \gg n_0, p_0$$

$$R_{Auger} = (C_n + C_p)(\Delta n)^3$$

and the effective lifetime becomes:

$$\tau_{Auger} = \frac{1}{(C_n + C_p)(\Delta n)^2}$$

- The recombination rate rises **very sharply** ($\propto \Delta n^3$) with injection level.
- The lifetime **drops** as injection increases ($\propto 1/\Delta n^2$).

Auger recombination Process

The **Auger lifetime** is inversely proportional to carrier density squared:

$$\tau_{Auger} = \frac{1}{C_n n^2 + C_p p^2}$$

Hence, lifetime decreases sharply as doping or injection increases.

Auger recombination converts the recombination energy of an electron–hole pair into **heat** by exciting a third carrier. It scales with the cube of carrier density and increases with temperature, making it the dominant loss mechanism in heavily doped or high-injection semiconductors.

Quantity	Trend with T	Explanation
Auger coefficients C_n, C_p	Increase with T (roughly $\propto T^2$ – T^3)	Higher lattice temperature broadens energy distributions; more carriers have enough energy to satisfy momentum & energy conservation → more probable three-particle collisions
Carrier lifetime τ_{Auger}	Decreases with T	Because C_n, C_p increase and intrinsic carrier density n_i also rises strongly with T
Overall effect	Faster Auger recombination at higher T	Enhanced phonon interactions and larger carrier populations